# Zachary A. Bolton.

Staff ML Engineer · AI Evaluation & Reliability

U.S. citizen · Based in Brazil · Remote preferred · Open to relocation

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> Staff ML engineer building real-time AI systems and the evaluations that govern their release. I've led engineering teams, trained speech models, and made eval-driven development a shared practice. Now bringing that experience into AI safety: behavioral evaluations, adversarial testing, and research infrastructure.

## Experience

**Boostlingo** Staff Machine Learning Engineer

August 2025–present · Promoted to Staff September 2026

- Lead engineer for a real-time AI interpreter; led four engineers. Built C#/.NET and Orleans voice infrastructure spanning voice-activity detection, conversation management, telephony, and rollout controls.
- Drove the organization's shift to eval-driven development through shared tooling, tech talks, and hands-on coaching. Built multilingual datasets, automated scoring, provenance records, and required CI gates; helped teams beyond the interpreter adopt the practice.
- Evaluated 29 languages and worked with language experts on release criteria, informing the rollout to 15 generally available languages by October 2026. Added independent language-ID checks after quality scores missed wrong-language output.
- Developed a modular intervention framework and a direct-address guard to keep the interpreter in role. Built spoken adversarial tests with ordinary-speech controls and detector-on/off comparisons.
- Fine-tuned a pyannote speech model on ~92 hours of staged conversations for crosstalk detection. Fixed evaluation and codec defects, tested on held-out meeting audio, and shipped detection in shadow mode, with live interruptions disabled.
- Designed, conducted, and authored a public medical AI interpreting accuracy study. Lead presenter for MT Evaluation for Spoken AI Interpreting at AMTA 2026.

**Kid Chess** Software Engineer & Director of Technology

2015–August 2025

- Moved 4,000 students online during the pandemic, helping retain over 90% of staff and keep the company afloat. Enabled a full return to in-person teaching plus a continuing online revenue stream.
- Grew engineering and QA from two to nine; hired, trained, and led delivery. Owned the online platform's product, engineering, and operations.
- Built LLM student-feedback reports with an anti-sycophancy stage and rubric-based human review to catch unsupported praise and guide improvements.

## Research & safety experiments

**University of London** Independent ML Research

April 2024–April 2025

- Built a Siamese SBERT authorship-verification pipeline in Python/PyTorch on AWS: genre-balanced data, text-distortion experiments, and external VALLA benchmarking. Earlier work explored stylometry and neural models.
- Investigated apparently perfect Brier scores: logged predictions, reran experiments, and traced the result to a scoring artefact. Documented methods and limitations in a public research report.
- **Independent safety experiments, AI-assisted:** built controlled résumé prompt-injection tests in Inspect, with visually identical PDFs, a clean control, and decision-based scoring. In a Petri behavioral-auditing work sample, separated attempted from completed actions and identified judge inconsistencies and simulator drift.
- **Harmful-request classification:** compared TF-IDF and DeBERTa, documenting false positives, generalization limits, and reproducibility. AI-assisted implementation and analysis.
- **Gender bias in chess:** analyzed discussions before and after *The Queen's Gambit*; documented sampling limits and how positive sentiment scores can miss sexist content.

## Education & tools

**BSc Computer Science, Machine Learning & AI** — University of London, First Class, 2025 · 90% average.

Python · PyTorch · Hugging Face · Inspect AI · C#/.NET · Orleans · Datadog LLM Observability · AWS · Git/CI

## AI safety engagement

Self-directed study since 2016, beginning with Robert Miles. BlueDot AGI Strategy, 2026: developing an eval proposal on whether AI notetakers favor the more powerful or fluent speaker. Intend to take Technical AI Safety next. DeepLearning.AI, *Red Teaming LLM Applications*, January 2026.